AgenticECG: Autonomous Hypothesis-to-Factor Discovery Framework for ECG Classification

Tianxiang Hao, Yu Wang, Chenxi Song, Kejun He, Ya Zhou, and Haohuan Fu, *Fellow, IEEE*

***Abstract*— Existing AI-ECG methods predominantly rely on end-to-end black-box models, while traditional feature-based approaches require manually designed and domain-specific ECG features. We propose AgenticECG, an autonomous hypothesis-to-factor discovery framework that leverages agents powered by large language models (LLMs) to automatically propose human-interpretable ECG factors for classification. Using these factors, we train an XGBoost classifier and evaluate it on two publicly available benchmarks for occlusion myocardial infarction (OMI) and low left ventricular ejection fraction (LVEF) detection against official deep learning baselines and human experts. AgenticECG achieves competitive or superior performance across both tasks, demonstrating its potential for interpretable ECG classification.**

***Clinical Relevance*—AgenticECG provides an automated approach for discovering human-interpretable ECG factors, potentially enabling more transparent and clinically meaningful ECG-based decision support.**

I. INTRODUCTION

Electrocardiography (ECG) is one of the most widely used tools for assessing cardiac health and detecting cardiovascular diseases. The rapid development of artificial intelligence (AI) has substantially improved the performance of automated ECG analysis, enabling the detection of various cardiovascular diseases directly from ECG signals. However, most existing AI-based ECG methods rely on end-to-end black-box models and therefore provide limited interpretability. In contrast, traditional feature-based methods offer more interpretable representations, but their performance depends heavily on manually designed features that require substantial domain expertise.

To address these limitations, we propose AgenticECG, an autonomous hypothesis-to-factor discovery framework that leverages large language model (LLM)-based agents to automatically propose ECG factors for classification.

II. METHODS

AgenticECG uses an iterative two-agent framework[1] based on DeepSeek-V4-Flash [2] (Figure 1). A doctor agent proposes interpretable hypotheses, while a disease-blind data-science agent converts each hypothesis into an executable binary or continuous factor using either the median beat or the full 12-lead ECG. Valid factors are added to the factor library for downstream XGBoost classification, while the test set is reserved for final evaluation.

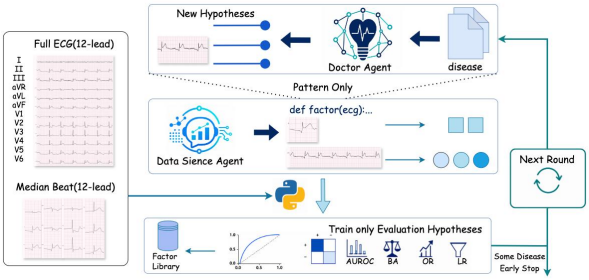


Figure 1. Overview of AgenticECG.

III. RESULTS

Based on the ECG factors discovered by AgenticECG, XGBoost models are trained on 17,960 and 72,475 ECGs for OMI and low-LVEF detection, respectively, and evaluated using the corresponding official protocols. It achieves the best OMI F1-score (43.2; Table I) and the best low-LVEF AUROC (85.7), AUPRC (60.5), and F1-score (59.9; Table II), outperforming the official deep learning baselines.

TABLE I. PERFORMANCE COMPARISON FOR OMI DETECTION

Method	Sensitivity	Specificity	PPV	NPV	Accuracy	F1
CNN Baseline*	**69.7**	87.3	27.7	**97.6**	86.1	39.6
Human Experts Mean*	35.3	**95.7**	**37.3**	95.5	**91.8**	35.4
AgenticECG(Ours)	57.8	92.5	34.4	97.0	90.2	**43.2**

Note: Our results were obtained from the official test website; other results (*) are from [3].

TABLE II. PERFORMANCE COMPARISON FOR LVEF DETECTION

Method	AUROC	AUPRC	F1
Columbia-mini model*	85.2 (83.9–86.5)	59.9 (56.6–63.3)	57.9 (55.4–60.3)
AgenticECG(Ours)	**85.7** (84.2–87.1)	**60.5** (57.4–63.8)	**59.9** (57.6–62.2)

Note: Evaluation metrics follow [4]; comparison results (*) are from [4].

Tianxiang Hao and Haohuan Fu are with the Shenzhen International Graduate School, Tsinghua University, Shenzhen, China. Yu Wang and Kejun He are with the Institute of Statistics and Big Data, Renmin University of China, Beijing, China. Chenxi Song is with the Department of Cardiology, Fuwai Hospital Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China. Ya Zhou is with the Department of Information Center, Fuwai Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China. **Correspondence: Ya Zhou (zhouya@fuwai.com) and Haohuan Fu (haohuan@tsinghua.edu.cn).*

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